

COAL
Assignment – 03
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L1F24BSCS0506

CODE

```
.model small
.stack 0x100
.data
    msg db 'Hey, I am Sharjeel Ahmad currently pursuing BsCs at University of
Central Punjab, Lahore. This is assignment - 03 of COAL...!$'
    row dw 24
    col dw 0
.code
main proc
    mov ax, @data
    mov ds, ax

    mov ax, offset msg
    push ax
    mov ax, row
    push ax
    mov ax, col
    push ax

    call printString
    add sp, 6
mov ax, 0x4c00
int 0x21
main endp
```

;function definition

printString proc

push bp

mov bp, sp

push ax

push bx

push cx

push dx

push si

push di

mov si, [bp + 4] ;offset string

mov ax, [bp + 6] ;row

mov bl, 80

mul bl ;ax = row * 80

add ax, [bp + 8] ;ax = row * 80 + col

shl ax, 1 ;ax = (row * 80 + col) * 2

mov di, ax ; di = video memory offset

mov ax, 0xB800 ;video memory segment

mov es, ax

printLoop:

mov al, [si]

cmp al, '\$'

```

        je exit

        mov es:[di + 0], al

        mov byte ptr es:[di + 1], 0xb3    ;grey character with black
background
        add si, 01
        add di, 02

        cmp di, 4000
        jb printLoop

        call scrollUpward

        mov di, 3840

        jmp printLoop
exit:
        pop di
        pop si
        pop dx
        pop cx
        pop bx
        pop ax
        pop bp
        ret

printString endp

;function definition
scrollUpward proc

```

push si

push di

push cx

push ax

mov ax, 0xB800

mov es, ax

mov si, 160 ;2 * 80

mov di, 0

mov cx, 1920 ;80 * 24

scrollLoop:

mov ax, es:[si]

mov es:[di], ax

add si, 2

add di, 2

loop scrollLoop

mov di, 3840

mov cx, 80

clearWithSpace:

mov es:[di], ' '

add di, 2

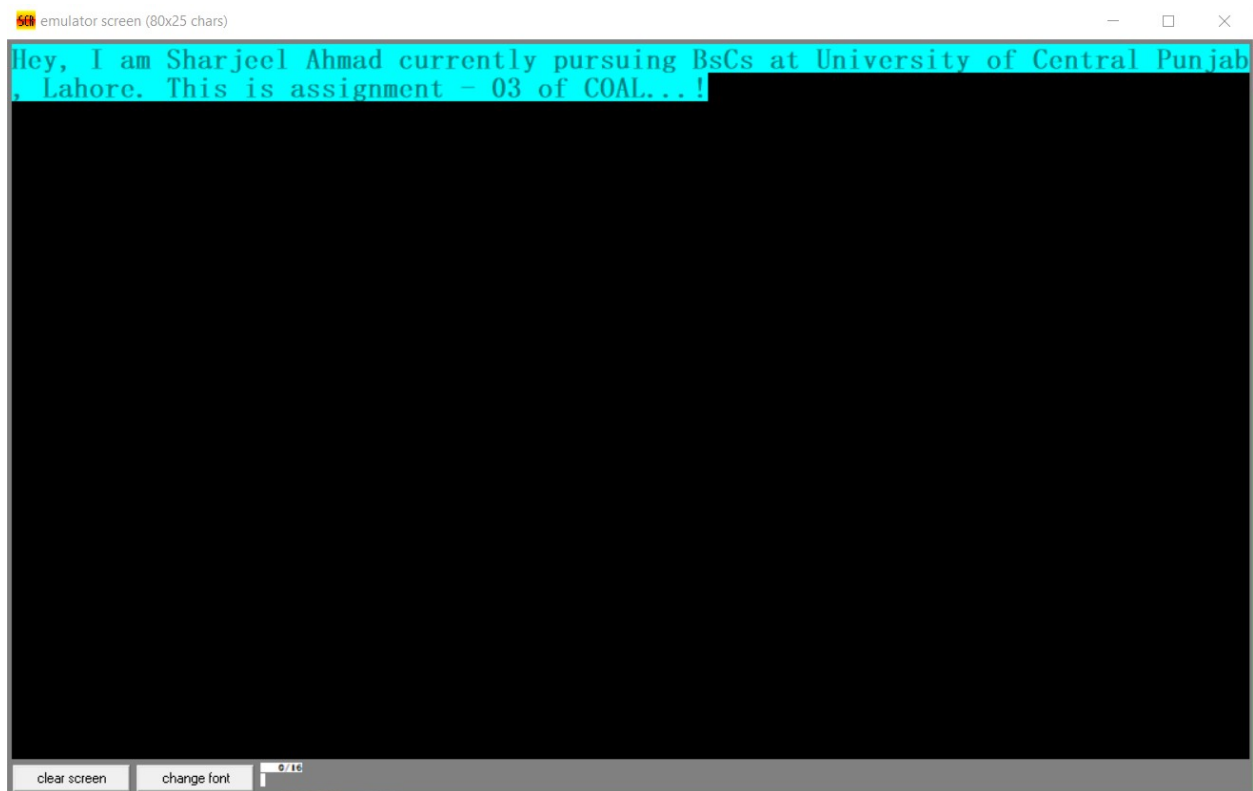
loop clearWithSpace

exitScroll:

```
    pop ax
    pop cx
    pop di
    pop si
ret
scrollUpward endp
```

Screen Shots

At $r = 0$, $c = 0$



At $r = 24$, $c = 0$

